

CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

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Case 18-2025: A 63-Year-Old Woman with Dyspnea on Exertion

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PRESENTATION OF CASE

Dr. Jacquelyn M. Nestor (Medicine): A 63-year-old woman was evaluated at this hospital because of dyspnea on exertion that had persisted for 2 months.

Fifteen months before the current presentation, the patient received a diagnosis of rheumatoid arthritis in the rheumatology clinic of another hospital after she had presented with a 6-month history of swelling and stiffness in the hand joints and skin tightening on several fingers. An evaluation revealed swan-neck deformities in both hands, and radiography showed marginal erosion of the hands. The white-cell count and differential count, the mean corpuscular volume, and the blood levels of thyrotropin, calcium, bilirubin, creatine kinase, total protein, globulin, and vitamins B₁₂ and D were normal. Antibodies to β_2 -glycoprotein, cardiolipin, thyroid peroxidase, double-stranded DNA, topoisomerase 1, mitochondria, smooth muscle, Smith antigen, ribonucleoprotein, Ro antigen, and La antigen were not detected. In addition, there were no antibodies to *Borrelia burgdorferi*, hepatitis B core antibodies, or hepatitis C antibodies, and an interferon- γ release assay for *Mycobacterium tuberculosis* was negative. Other laboratory test results are shown in Table 1. Treatment with methotrexate was initiated, and the joint swelling reduced during the next 2 months.

Approximately 1 year later, and 7 weeks before the current presentation, exertional dyspnea and fatigue developed. The patient first noticed that she was “struggling” on hikes with her friends, whereas she had often been the first to reach the end of the hiking trail. During the next 2 weeks, dry cough developed, and dyspnea progressed to the point that the patient needed to rest after 10 minutes of routine treadmill and elliptical exercises. She started to use three pillows to sleep, and she noted abdominal distention, anorexia, and postprandial nausea.

One month before the current presentation, the patient was evaluated by her primary care physician, affiliated with the other hospital. Screening for severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infection was negative. The results of urinalysis and the blood levels of thyrotropin, amylase, lipase, and

Author affiliations are listed at the end of the article.

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CME



Table 1. Laboratory Data.*

Variable	Reference Range, Adults, Other Hospital	15 Mo before Current Admission, Outpatient Rheumatology Clinic	1 Mo before Current Admission, Primary Care Clinic	Reference Range, Adults, This Hospital†	On Current Admission, This Hospital
Erythrocyte sedimentation rate (mm/hr)	0–20	55	—	0–20	70
C-reactive protein (mg/liter)	<8.0	25.2	—	<8.0	47.1
High-sensitivity C-reactive protein (mg/liter)	0–3.00	—	40.74	—	—
Antinuclear antibody	Negative	Positive at 1:80, homogeneous	—	Negative	Positive at 1:640, homogeneous
Rheumatoid factor (IU/ml)	Negative	Positive	—	0–14	57
Anti-citrullinated cyclic peptide (U/ml)	<20	>250	—	0–16	>500
Hemoglobin (g/dl)	12.0–16.0	12.5	11.1	12.0–16.0	9.8
Hematocrit (%)	37.0–47.0	36.6	36.1	36.0–46.0	31.4
White-cell count (per μ l)	4500–10,500	5800	7700	4500–11,000	7150
Platelet count (per μ l)	150,000–450,000	312,000	383,000	150,000–400,000	346,000
Sodium (mmol/liter)	136.0–146.0	139.4	137.9	135–145	139
Potassium (mmol/liter)	3.5–4.0	4.1	4.0	3.4–5.0	4.9
Chloride (mmol/liter)	97.0–110.0	103.6	106.8	98–108	105
Carbon dioxide (mmol/liter)	23.0–29.0	28.6	30.6	23–32	23
Urea nitrogen (mg/dl)	7–27	8	4	8–25	13
Creatinine (mg/dl)	0.60–1.50	0.50	0.60	0.60–1.50	0.50
Glucose (mg/dl)	70–100	166	130	70–110	104
Albumin (g/dl)	3.5–5.0	3.8	3.0	3.3–5.0	3.0
Alanine aminotransferase (U/liter)	8–23	<5	12	7–33	10
Aspartate aminotransferase (U/liter)	7–35	<7	13	9–32	41
Alkaline phosphatase (U/liter)	98	42–98	100	45–115	109
N-terminal pro-B-type natriuretic peptide (pg/ml)	<900	—	389	<900	672

* To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for glucose to millimoles per liter, multiply by 0.05551.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

γ -glutamyltransferase were normal; other laboratory test results are shown in Table 1. Chest radiography reportedly showed diffuse reticular interstitial markings, and abdominal ultrasonography reportedly showed hepatic cysts, pleural fluid, and hydronephrosis on both sides.

The following week, the patient was evaluated by a cardiologist affiliated with the other hospital. An electrocardiogram (ECG) reportedly showed ectopic atrial rhythm, T-wave inversions in infe-

rior and precordial leads, and prolongation of the corrected QT interval (QTc). Transthoracic echocardiography (TTE) reportedly showed normal biventricular size and systolic function, with no evidence of hemodynamically significant valvular disease. However, the study was notable for an abnormal echogenic structure adjacent to the aortic root that appeared to be most consistent with soft tissue, as well as fluid and diffuse echogenic material in the pericardial space.

Computed tomography (CT) of the chest, performed without the administration of intravenous contrast material, reportedly showed pericardial fluid and thickening, pleural fluid on the left side, pararenal fluid on both sides, and infiltrative changes in the fat in the mediastinum and surrounding the aorta.

Treatment with methotrexate was discontinued, and colchicine and furosemide were started. Owing to persistent symptoms, the patient took time off from work. One week later, limited echocardiography reportedly showed that the pericardial effusion had decreased in size, but

organized intrapericardial material was again noted. The patient was evaluated by her rheumatologist, and the dose of furosemide was increased because of leg edema and persistent dyspnea.

Dr. Reece J. Goiffon: Eight days before the current presentation, CT of the abdomen and pelvis (Fig. 1A) showed anterior pararenal and bilateral perirenal fluid and soft tissue, as well as bilateral pleural effusions. A small pericardial effusion, hepatic cysts, and a uterine leiomyoma were noted. One week later, abdominal magnetic resonance imaging (MRI) was performed.

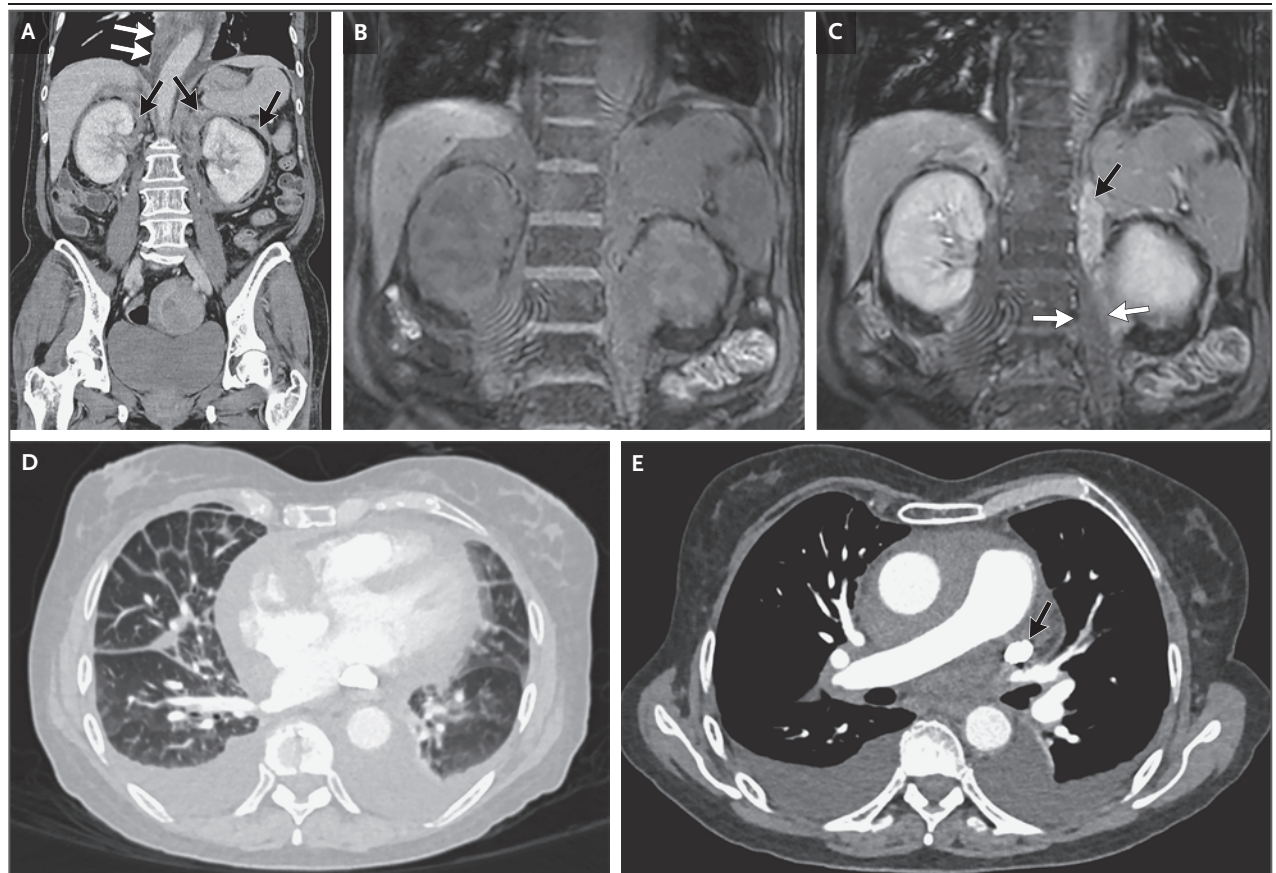


Figure 1. Imaging Studies.

A coronal CT image obtained after the administration of intravenous contrast material (Panel A) shows enhancement of soft tissue in the retroperitoneal space (black arrows). Abnormal tissue is also present in the partially visible lower mediastinum (white arrows). No ascites or other related abnormality is shown. T1-weighted coronal MRIs obtained before and after the administration of intravenous contrast material (Panels B and C, respectively) show enhancement of soft tissue in the lower mediastinal and retroperitoneal spaces (Panel C, black arrow) as compared with the psoas muscle (Panel C, white arrows). An axial CT image obtained after the administration of intravenous contrast material in lung windows (Panel D) shows bilateral pleural effusions, interstitial pulmonary edema, and pericardial thickening. An axial CT image obtained in soft-tissue windows at the level of the pulmonary artery (Panel E) shows soft-tissue thickening throughout the mediastinum, surrounding the aorta and pulmonary artery; a duplicate left superior vena cava is present (arrow).

T1-weighted imaging (Fig. 1B and 1C) showed extensive abnormal hypointense soft-tissue thickening in the retroperitoneal space surrounding both kidneys, the abdominal aorta, and the descending thoracic aorta. The head of the pancreas had a bulky appearance. Pleural effusions and pericardial thickening were noted.

Dr. Nestor: The next day, dyspnea worsened, and the patient was referred to the emergency department of this hospital. In the emergency department, review of systems was notable for persistent dyspnea, which increased with minimal activity and had prevented the patient from sleeping for more than 3 hours at a time. The patient also reported leg edema, abdominal distention, and dry cough that was worse when she was in the supine position. She had dry mouth and polydipsia that had persisted for 2 months, along with polyuria that was characterized by colorless urine. The patient also noted a several-month history of “fast heart rates” and the perception that her heart was beating forcefully. Four weeks before the current presentation, the patient had sustained a fracture of the right distal radius after a fall that had occurred while she was cleaning, and the arm was immobilized in a cast. The degree of swelling and stiffness in the hands was similar to that seen 1 year earlier, after she had started treatment with methotrexate. She reported no chest pain or pressure, lightheadedness, syncope, fever, chills, night sweats, productive cough, weight change, emesis, abdominal pain, back or flank pain, hematuria or dysuria, arthralgia, rash, photosensitivity, hair or nail changes, or neurologic symptoms.

Other medical history was notable for surgical resection of a meningioma, a positive tuberculin skin test but a negative interferon- γ release assay 1 year earlier, a solitary lung nodule, hypertension, dyslipidemia, diabetes mellitus, *Helicobacter pylori* gastritis, adhesive capsulitis (frozen shoulder) for which the patient had received glucocorticoid injections, and knee arthroscopy. Medications included metformin, as well as calcium and multivitamin supplements; methotrexate, lisinopril, and atorvastatin had recently been discontinued. Colchicine had caused diarrhea; there were no other known adverse reactions to medications. The patient had completed vaccinations against SARS-CoV-2 approximately 7 months before the

current presentation. She was born and raised in East Asia. She worked in an office setting and lived with her husband in a wooded area of New England. She drank three glasses of wine per week and did not use tobacco or other substances. Her family history included coronary artery disease and colon cancer in her father, systemic lupus erythematosus in her son, and hypertension and diabetes mellitus in multiple relatives.

On examination, the temporal temperature was 36.4°C, the heart rate 77 beats per minute, the blood pressure 144/74 mm Hg, and the oxygen saturation 99% while the patient was breathing ambient air. The weight was 62.6 kg, the height 168 cm, and the body-mass index (the weight in kilograms divided by the square of the height in meters) 22.3. The patient appeared fatigued and was coughing during the examination. The mucous membranes were dry and cracked. The jugular venous pressure was 14 cm of water. Auscultation of the chest revealed diminished breath sounds and crackles in both lungs. There was abdominal distention but no tenderness or organomegaly. The right forearm was in a cast; distal hand strength and pulses were intact. Swan-neck deformities were present, but there were no other joint abnormalities. Edema was present in the lower legs. Skin tightening was seen on the fingers, but no rashes were identified.

The results of urinalysis, the white-cell count and differential count, the venous blood pH, and the blood levels of high-sensitivity troponin T, calcium, magnesium, phosphorus, bilirubin, and lactate dehydrogenase were normal; other laboratory test results are shown in Table 1. Screening for SARS-CoV-2 infection was negative.

Dr. Carola A. Maraboto Gonzalez: ECG revealed probable ectopic atrial rhythm, right-axis deviation, diffuse T-wave inversions in inferior and precordial leads, and QTc prolongation of 509 msec.

Dr. Goiffon: Chest radiography showed an enlarged cardiac silhouette, diffuse pulmonary edema, and small bilateral pleural effusions. CT of the chest (Fig. 1D and 1E), performed after the administration of intravenous contrast material, showed pulmonary edema with bilateral pleural effusions, a small pericardial effusion with exten-

sive pericardial thickening, and soft-tissue thickening surrounding the ascending aorta, the descending thoracic aorta, and the portions of the kidneys that were visible on imaging.

Dr. Nestor: Furosemide was administered intravenously. The patient was admitted to the hospital. On the second hospital day, the results of serum protein electrophoresis and the levels of IgG subclasses were normal.

Dr. Maraboto Gonzalez: On the third hospital day, TTE showed diffuse echogenic material in the pericardial space, pleural effusions, and an atypical appearance of tissue around the aortic root (Fig. 2A, 2B, and 2C and Video 1).

Dr. Goiffon: Cardiac MRI was also performed. Abnormal gadolinium enhancement in the pericardium and soft-tissue edema in the mediasti-

num were noted, with neither finding affecting the myocardium (Fig. 2D and 2E).

Dr. Nestor: Diagnostic tests were performed.

DIFFERENTIAL DIAGNOSIS

Dr. Malissa J. Wood: This 63-year-old woman with hypertension, diabetes, and rheumatoid arthritis presented with a 2-month history of dyspnea on exertion. She was found to have an elevated jugular venous pressure, diminished breath sounds and crackles in both lungs, abdominal distention, and leg edema — findings suggestive of heart failure. Echocardiography showed a normal left ventricular ejection fraction and no evidence of hemodynamically significant valvular disease. The patient's clinical syndrome could be consistent



A video showing
TTE is available
at [NEJM.org](https://www.nejm.org)

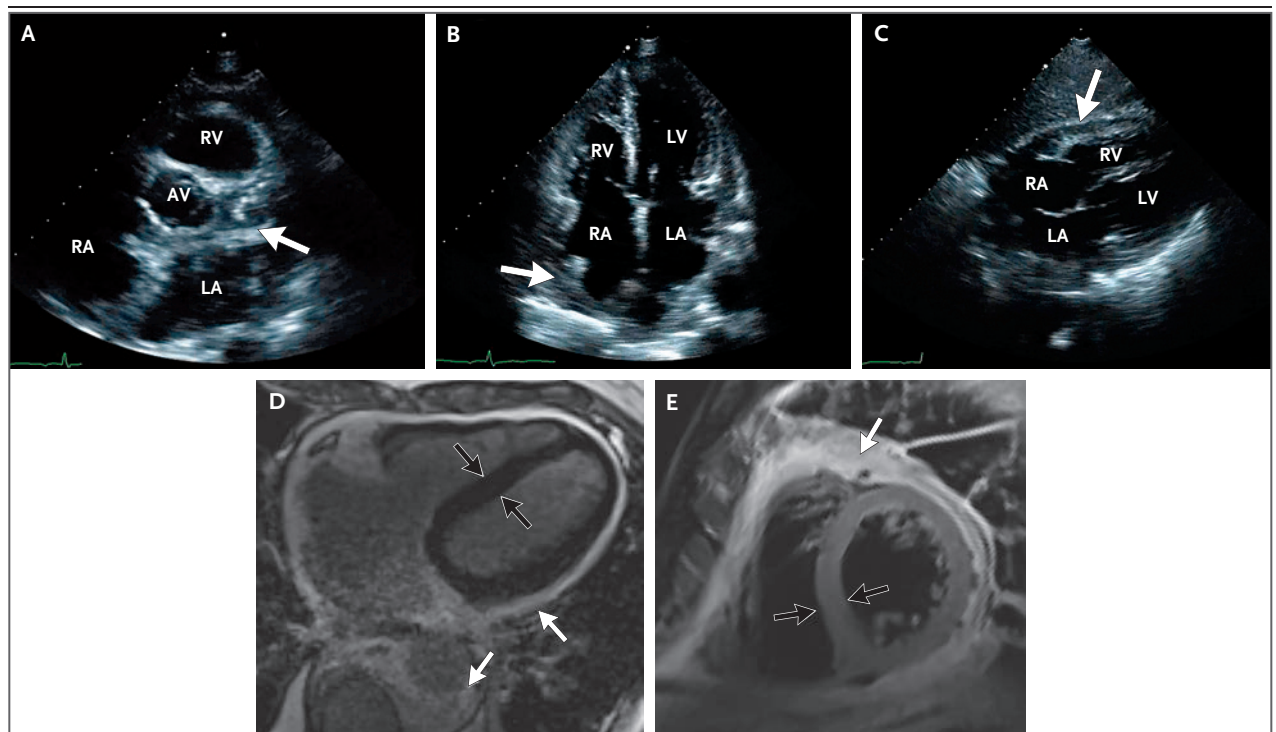


Figure 2. Cardiac Imaging Studies.

A transthoracic echocardiogram (TTE) obtained in the parasternal short-axis view at the level of the aortic valve (Panel A) shows an irregular and nonmobile echogenic structure, measuring approximately 2 cm in thickness, adjacent to the aortic root (arrow), with an echogenicity that is most consistent with soft tissue. TTEs obtained in the apical four-chamber view (Panel B) and in the subcostal view (Panel C) show echogenic material within the pericardial space (arrows). AV denotes aortic valve, LA left atrium, LV left ventricle, RA right atrium, and RV right ventricle. A cardiac MRI obtained in the four-chamber view (Panel D) shows late gadolinium enhancement of the mediastinal and pericardial tissue (white arrows) but not of the myocardium (black arrows). A T2-weighted cardiac MRI obtained in the short-axis view (Panel E) shows edema and pericardial thickening (white arrow) and surrounding pulmonary edema but a normal myocardial signal (black arrows).

with heart failure with preserved ejection fraction. However, imaging studies showed nodular soft tissue in the pericardial, periaortic, pararenal, and retroperitoneal spaces. These unusual findings, in addition to elevated levels of inflammatory markers and the 2-month history of polydipsia and polyuria, suggest a systemic process with secondary involvement of the heart. In working toward the most likely diagnosis, I will consider infections, cancer, and inflammatory diseases.

INFECTION

Infection is an important consideration in this patient, who had been receiving immunosuppressive therapy with methotrexate for rheumatoid arthritis. Infectious aortitis due to tuberculosis, Lyme disease, or a disseminated bacterial infection could explain diffuse involvement of the pericardium, aorta, and retroperitoneum. The patient was born and raised in East Asia, an area of the world with a high burden of tuberculosis. However, the negative interferon- γ release assay 1 year before this presentation makes tuberculosis unlikely. Antibodies to borrelia were not detected the year before this presentation. However, the patient lived in a wooded area of New England and therefore had ongoing risk for the acquisition of Lyme disease. The absence of fever and leukocytosis make a disseminated bacterial infection unlikely, although her use of immunosuppressive therapy remains a consideration.

CANCER

The diffuse, extensive soft-tissue deposition could be consistent with lymphoma or widely metastatic cancer. The patient had not previously received a diagnosis of cancer and did not report fever, weight loss, or night sweats. No lymphadenopathy was noted on examination or on any of the imaging studies. The white-cell count and platelet count were normal. Although cancer remains on the differential diagnosis, I will consider other possible causes of her presentation.

INFLAMMATORY DISEASE

The patient had received a diagnosis of rheumatoid arthritis 15 months before the current presentation, when she presented with a pattern of

joint involvement typical of rheumatoid arthritis, swan-neck deformities, bone erosions on plain films of the hands, and elevated levels of both rheumatoid factor and anti-citrullinated cyclic peptide. Rheumatoid vasculitis is an important consideration, given the presence of pericardial and aortic involvement. However, the extensive extracardiac soft-tissue thickening would be atypical for rheumatoid vasculitis. The absence of oral and genital ulcerations and of ocular disease makes a diagnosis of Behçet's syndrome-associated vasculitis unlikely. Imaging showed periaortic wall thickening (as opposed to aortic wall thickening), which makes aortitis unlikely, and the absence of extensive involvement of vessels other than the aorta makes vasculitis an unlikely diagnosis overall. The presence of soft-tissue thickening in the retroperitoneal and pararenal spaces and around the pancreatic head suggests the possibility of less common inflammatory diseases: IgG4-related disease and Erdheim-Chester disease.^{1,2}

IGG4-RELATED DISEASE

IgG4-related disease is an immune-mediated fibroinflammatory condition that affects multiple organs. The disease is more common in men than in women (3:1 ratio). Inflammatory markers such as the C-reactive protein level and the erythrocyte sedimentation rate are typically elevated, as seen in this patient. The serum IgG4 level is elevated in more than 70% of cases; the presence of an IgG4 level of more than twice the upper limit of the normal range (reference value, <140 mg per deciliter) is 99% specific for IgG4-related disease. Although the IgG4 level in this patient was not provided, even a normal level would not rule out IgG4-related disease. Diagnosis requires pathological confirmation. Histologic features include the following: a lymphoplasmacytic infiltrate with IgG4+ plasma cells arranged in a storiform pattern of fibrosis; a ratio of IgG4+ plasma cells to IgG+ plasma cells of more than 40%; associated fibrosis; and evidence of obliterative phlebitis.

IgG4-related disease can lead to soft-tissue deposition in the pericardium, and this patient had a presenting clinical syndrome that was consistent with heart failure with preserved ejection fraction, which could be explained by func-

tional constrictive physiological features due to pericardial thickening. Additional features in this patient that are consistent with a diagnosis of IgG4-related disease include retroperitoneal fibrosis leading to hydronephrosis, possible involvement of both the pleura and the pancreatic head, and circumferential periaortitis.³ In patients with IgG4-related disease, perivascular disease (periaortitis) occurs more frequently than primary involvement of the vessel wall. When the vessel wall is directly involved, complications can include aneurysm, dissection, and vessel rupture. In this patient, imaging showed evidence of infrarenal periaortitis that was extending inferiorly with involvement of the iliac vessels, a pattern that is strongly suggestive of IgG4-related retroperitoneal fibrosis. However, there was also involvement of the aortic root and ascending aorta, which would be atypical for IgG4-related disease, as would the absence of lymphadenopathy and eosinophilia.

ERDHEIM–CHESTER DISEASE

Erdheim–Chester disease is a rare non–Langerhans’-cell histiocytic multisystem disorder.⁴ Bone involvement is present in up to 74% of cases and can include osteosclerosis of the long bones such as the femur and tibia. This patient did not report long-bone pain; none of the imaging studies examined the long bones, so bony involvement remains a possibility. Similar to IgG4-related disease, Erdheim–Chester disease is often associated with an elevated C-reactive protein level and erythrocyte sedimentation rate.⁵ Histologic features include sheets of foamy histiocytes, with or without histiocytic infiltration of extraosseous tissues. Although the diagnosis of Erdheim–Chester disease is established by means of tissue biopsy, findings on imaging can provide strong supportive evidence.

Affected patients commonly have neurologic involvement, including central diabetes insipidus, which could explain the patient’s 2-month history of polydipsia and polyuria. She did not have evidence of other neurologic involvement such as visual disturbance, ataxia, stroke, or headache. Retroperitoneal fibrosis leading to hydronephrosis may occur in patients with Erdheim–Chester disease. The presence of perirenal nodular soft-tissue deposition in a ring pattern (known as

“hairy kidney”) is highly suggestive of this diagnosis. Pulmonary involvement, including pleural effusions, can also occur.

Cardiovascular involvement occurs in approximately one third of patients with Erdheim–Chester disease. Manifestations include coronary artery disease (in 25% of patients), myocardial infiltration (in 25%), aortic and vascular disease (in 17%), pericardial involvement (in 13%), conduction disturbance (in 8%), and endocardial involvement (in 4%).⁶ Aortic and vascular disease is caused by adventitial and periadventitial infiltration, which results in a “coated” appearance on imaging. This patient reported symptoms that were possibly consistent with dysrhythmia, and ECG showed QTc prolongation. She also had evidence on imaging of periaortic involvement, a pericardial effusion with pericardial thickening, and circumferential soft-tissue coating of the thoracic and abdominal aorta.

This patient’s presentation is most consistent with Erdheim–Chester disease.⁶ To establish this diagnosis, I would perform a biopsy of the most accessible area of fibrosis.

CLINICAL IMPRESSION

Dr. Nestor: In this patient, the diffuse soft-tissue thickening in the retroperitoneum and surrounding the large vessels was suggestive of IgG4-related disease or Erdheim–Chester disease. Although diabetes insipidus can occur with both diseases, it is more common with Erdheim–Chester disease. After consultation with colleagues from endocrinology, we confirmed the presence of central diabetes insipidus, and treatment with desmopressin was initiated. We obtained levels of IgG subclasses, which were normal. In differentiating between Erdheim–Chester disease and IgG4-related disease, tissue biopsy is the most definitive approach, but Erdheim–Chester disease can be further characterized by fluorodeoxyglucose (FDG) uptake in the long bones on positron-emission tomography (PET), a finding that is not seen with IgG4-related disease.

CLINICAL DIAGNOSIS

Erdheim–Chester disease.

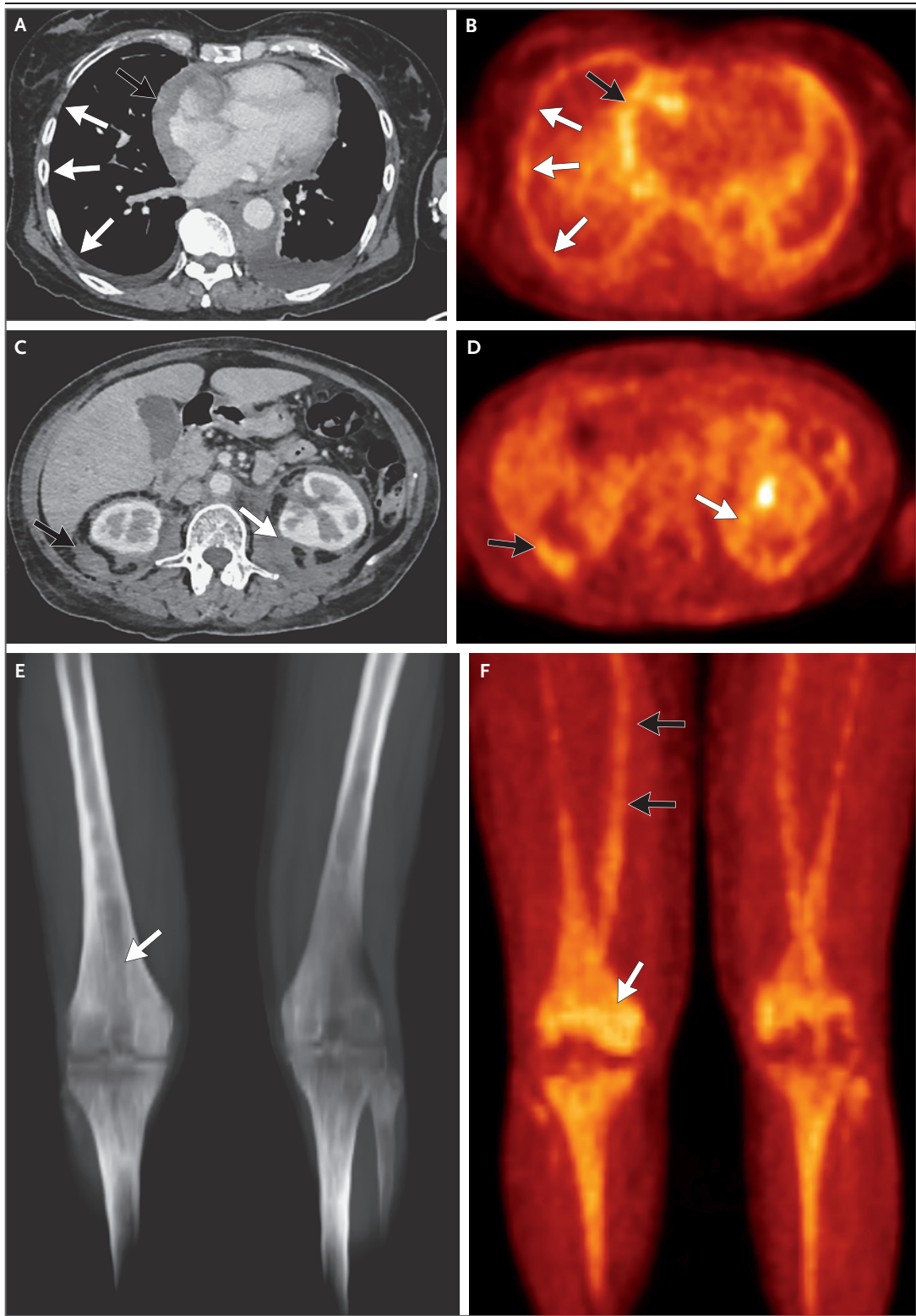


Figure 3 (facing page). PET and CT Imaging Studies.

A CT image (Panel A) shows soft-tissue thickening in the pericardium (black arrow) and pleura (white arrows). A positron-emission tomographic (PET) image (Panel B) shows corresponding increased fluorodeoxyglucose (FDG) uptake in the pericardium (black arrow) and pleura (white arrows). Additional CT and PET images (Panels C and D, respectively) show soft-tissue thickening and corresponding increased FDG uptake in the posterior pararenal space (black arrows) and perirenal space (white arrows). An average-intensity-projection coronal CT image of the knees (Panel E) shows increased sclerosis of the medullary cavity (arrow) that corresponds to intense FDG uptake on a PET image (Panel F, white arrow). Abnormal soft tissue around the femoral arteries is also associated with FDG uptake (Panel F, black arrows).

DR. MALISSA J. WOOD'S DIAGNOSIS

Erdheim–Chester disease.

DIAGNOSTIC TESTING

Dr. Goiffon: PET-CT showed intense FDG uptake in the soft tissue encasing the large mediastinal vessels, pericardium, and retroperitoneal compartments, as well as the pleural surfaces (Fig. 3A through 3D). CT of the lower legs showed abnormal enhancing soft-tissue thickening around both femoral arteries that extended to the popliteal arteries. There was diffuse sclerosis of the medullary cavities of the distal femora and tibiae (Fig. 3E). Concurrent PET showed increased FDG uptake around the femoral arteries and in the abnormally sclerotic bone marrow around the knees (Fig. 3F).

Dr. Bailey M. Hutchison: A core biopsy of the left retroperitoneal soft-tissue lesion was performed (Fig. 4). The lesion was composed primarily of atypical histiocytoid cells with small ovoid nuclei and abundant cytoplasm with features ranging from flocculent and eosinophilic to clear and foamy. Thin-walled vessels and small lymphocytes were scattered among the histiocytoid cells; no giant cells were seen. On the basis of the patient's clinical history and the histologic features, the differential diagnosis included histiocytoid neoplasms such as Langerhans'-cell histiocytosis and Erdheim–Chester disease, as well

as granulomatous processes such as fungal or mycobacterial infection.

Immunohistochemical staining of the biopsy specimen revealed that the lesional cells were positive for the histiocytic markers CD68 and CD163, as well as for PU.1 and factor XIIIa. The cells were negative for S100 (which ruled out Rosai–Dorfman disease), CD1a, langerin (which ruled out Langerhans'-cell histiocytosis), and pancytokeratin MNF116 (which ruled out epithelial neoplasm). Silver-based staining did not show any fungal elements within the lesion, and staining for acid-fast bacilli was negative. Mutation-specific immunostaining for BRAF V600E was positive, a finding that suggested the presence of this mutation within the lesion and made Erdheim–Chester disease the most likely diagnosis.⁷ A rapid molecular BRAF assay performed on formalin-fixed paraffin-embedded biopsy tissue confirmed the presence of a BRAF V600E/D mutation, which further supported the diagnosis of Erdheim–Chester disease.

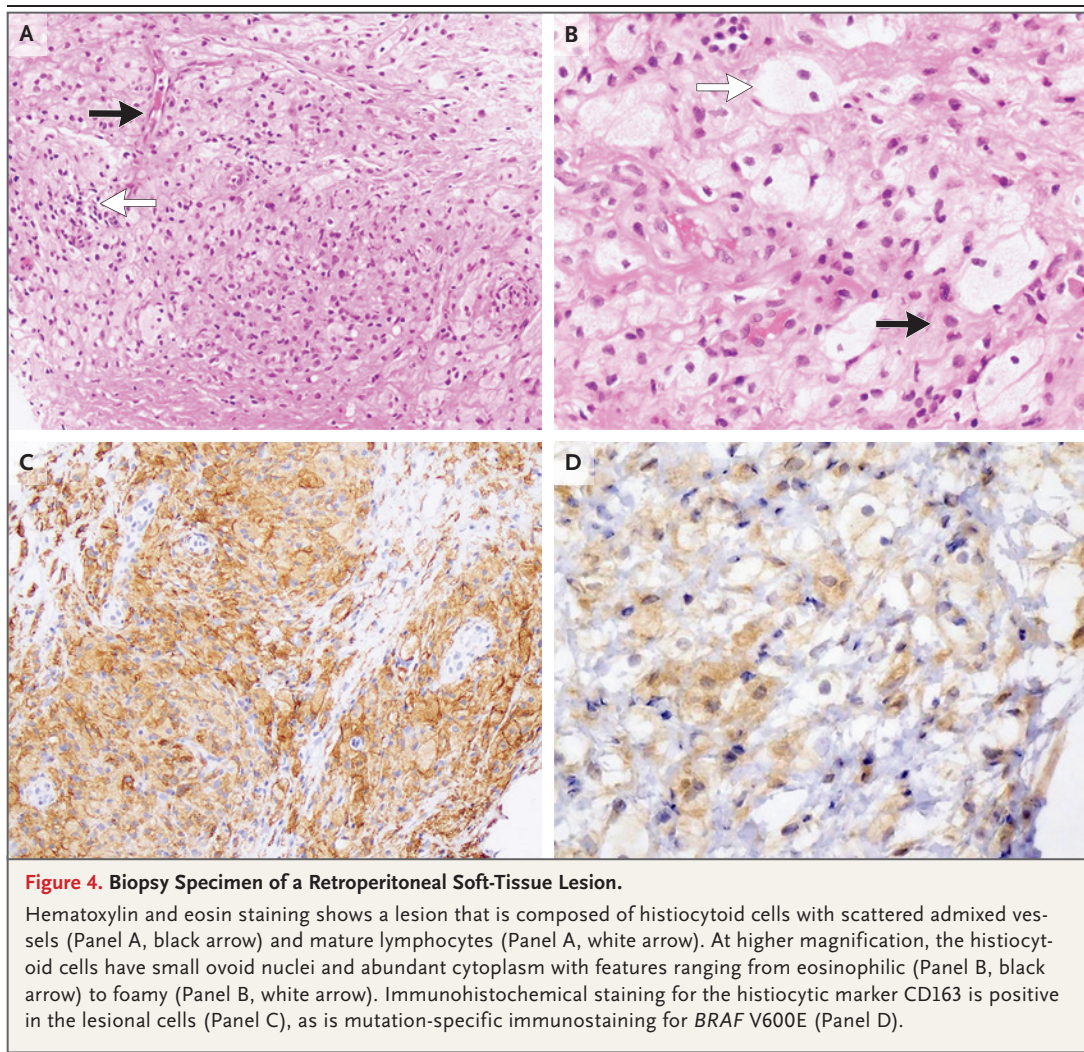
PATHOLOGICAL DIAGNOSIS

Erdheim–Chester disease.

DISCUSSION OF MANAGEMENT

Dr. Eric D. Jacobsen: Management of Erdheim–Chester disease is hampered by the rarity of the disease and the lack of data from randomized trials. Erdheim–Chester disease is a chronic disease, although spontaneous improvement and remissions have been known to occur. The safety of close observation as compared with immediate treatment is unclear, but a strategy of observation can be safe in some patients with asymptomatic disease and without critical organ involvement (e.g., neurologic, cardiovascular, or renal involvement). In the past, the most common treatments were interferon-alpha and pegylated interferon, which have been associated with prolonged survival.⁸ However, the side effects of interferon treatment can adversely affect quality of life.

A major therapeutic advance in the treatment of Erdheim–Chester disease was the development of molecularly targeted therapy with vemurafenib



in patients with *BRAF* V600E, a mutation that occurs in approximately 50% of cases.⁷ In a study of vemurafenib involving 22 patients with *BRAF* V600E-mutated Erdheim–Chester disease, approximately 55% of the patients had a response; among the 12 patients in whom response was assessed with PET, the response rate was 100%. Adverse cutaneous reactions (e.g., squamous-cell carcinomas) and cardiovascular side effects (e.g., hypertension and QTc prolongation) were common despite dose reductions.^{9–11}

The appropriate dose and duration of vemurafenib treatment have not been elucidated. In a large series involving patients with Erdheim–Chester disease who were treated with targeted therapy, 20 patients discontinued treatment with a *BRAF* inhibitor at a median of 20 months; 75% of the patients who discontinued treatment had

disease progression within 6 months, although the 10 patients who resumed treatment subsequently had a response, which suggested that there was no inducible resistance.¹² In the absence of further data, the duration of treatment with a *BRAF* inhibitor for Erdheim–Chester disease is considered to be indefinite, barring disease progression or unacceptable toxic effects. The duration is indefinite because Erdheim–Chester disease is thought to be chronic and incurable, and the goal of treatment is remission and disease control. Acquired resistance to *BRAF* inhibitors can occur but is uncommon.

BRAF wild-type Erdheim–Chester disease, which is not responsive to *BRAF* inhibitors, usually harbors a mutation in the mitogen-activated protein kinase pathway that can be targeted by MEK inhibitors. In a study of the MEK1/2 inhibitor

cobimetinib involving 12 patients with Erdheim–Chester disease, 89% of the patients had a response, with 100% of the responses ongoing at 1 year.¹³ Responses occurred regardless of the underlying mutation status, which could include *BRAF* V600E. The dose of cobimetinib was reduced in 56% of the patients, including 5 patients who had a decrease in the left ventricular ejection fraction.

Questions remain regarding the use of MEK inhibitors for the treatment of Erdheim–Chester disease, including the appropriate dose and duration of treatment. Whether MEK inhibitors are preferred over *BRAF* inhibitors, even in patients whose disease harbors *BRAF* V600E or another *BRAF* mutation that is sensitive to *BRAF* inhibition, remains unclear, as does the preferred sequence of treatment with *BRAF* and MEK inhibitors. Some reports suggest that MEK inhibitors are active in Erdheim–Chester disease with resistance to *BRAF* inhibitors.¹⁴ Whether the administration of *BRAF* inhibitors in combination with MEK inhibitors would improve efficacy and mitigate toxic effects remains unknown.^{15–17}

In this patient, indefinite first-line treatment with vemurafenib would have been appropriate, given the presence of *BRAF* V600E. However, continued QTc prolongation prohibited treatment with vemurafenib. Therefore, treatment with cobimetinib was initiated.

We expected that the patient's rheumatoid arthritis symptoms would abate after she started treatment for Erdheim–Chester disease. A large percentage of patients with concurrent clinically apparent autoimmune illness have relief of signs and symptoms of those diseases while receiving

treatment for Erdheim–Chester disease, particularly *BRAF* or MEK inhibitors.¹⁸

Dr. Nestor: Within a month after cobimetinib therapy was initiated, the patient was able to resume exercising. Treatment with diuretic agents was successfully stopped without a recurrence of leg edema, and treatment with desmopressin was discontinued without a recurrence of polydipsia or polyuria. After 3 months of treatment with cobimetinib, repeat PET showed decreased FDG avidity in the bones, pleura, and pancreas. Imaging also showed a decrease in the soft-tissue thickening surrounding the aorta, heart, and kidneys and within the retroperitoneum. Two years after the diagnosis, the patient was still taking cobimetinib, and PET showed only minimal residual FDG avidity, a finding consistent with regression of Erdheim–Chester disease. The joint pain had not recurred, and she had not received additional treatment with methotrexate for rheumatoid arthritis.

FINAL DIAGNOSIS

Erdheim–Chester disease.

This case was presented at the Harvard Medical School postgraduate course “Advances in Rheumatology,” directed by Dr. John H. Stone.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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